Amanda C. Ross, M.S.

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EDUCATION

University at Buffalo, SUNY — Ph.D. Computer Science & Engineering
Fall 2026 – Present | Advisor: Dr. Wenyao Xu, Embedded Sensing and Computing (ESC) Lab

San Jose State University — M.S. Biomedical Engineering
December 2021 | GPA: 3.5
Relevant Coursework: Medical Device Principles, Orthotics & Prosthetics, Biomedical Manufacturing, Regulatory & Clinical Affairs, Design Methods, Physiology for Engineers

University of California, Santa Cruz — B.A. Film & Digital Media | B.A. Visual Arts
Minor: Technology & Information Management
June 2015 | GPA: 3.6

EXPERIENCE

Freelance CAD Engineer – *Various Clients, San Jose, CA*
October 2023 – Present

- Delivered custom CAD designs and 3D printed prototypes for clients in consumer goods.
- Translated client concepts into 3D models, resolving form and function issues through iterative prototyping.
- Applied DFM principles, validated performance through hands-on testing, and refined designs based on user feedback.

Product Development Engineer – *Nordson Medical, San Jose, CA*
March 2022 – June 2023

- Assisted in execution of 5+ client medical device programs from concept through verification.
- Integrated customer research and clinical needs into product requirements and risk mitigation.
- Mentored summer interns in quality documentation and protocol development.
- Received Achieve and Inspire Awards for contributions to NPI and team leadership.

R&D Engineer – *NoLapse Pediatric Device Project (with UCSF surgeons)*
August 2020 – March 2022

- Co-developed an anti-stomal prolapse device in collaboration with UCSF pediatric surgeons.
- Secured over $130K in grants (STTR and Level-Up) and authored a 70-page thesis.
- Assisted in prototyping, animal studies, team management, and quality documentation.

Prototype Engineer – *Tortoise Beak Prosthesis Project, San Jose State University*
September 2021 - March 2022

- Designed a custom, two-piece, UV-resistant prosthetic for a Texas tortoise with chronic beak malocclusion.
- Used 3D printing, biocompatible materials, and magnetic components for field-ready application.
- Balanced anatomical constraints and caregiver usability in collaboration with caretakers and faculty.

Media Director – *David Cohen for San Jose City Council, District 4*
October 2019 – March 2020

- Produced campaign videos, promotional photography, and website contributing to a successful election.
- Built visual identity and handled multimedia strategy.

Research Associate – *Eisen Lab, UC Berkeley (E3S TTE REU)*
June 2018 – August 2018

- Conducted foundational research on intrinsically disordered transcription factors.
- Presented at SACNAS, the largest national conference supporting diversity in STEM.

Vice President – *SJCC Associated Student Government*
May 2018 – June 2019

- Led marketing and student access initiatives, including the $1 Tacos Affordable Food Initiative and Gym Access for All campaign.

Vice President – *SJCC Robotics & Automation Club*
September 2017 – June 2018

- Directed club programs such as 3D bioprinting workshops, robotics competitions, and AI forums.

Other Experience

- Specimen Processor – *Bioreference Laboratories*, 2018–2019
- Product Ambassador & Engineering Intern – *SE3D, Bioprinting for Education*, 2017
- Photography Intern – *Museum of Art & History*, 2016
- Academic Coach & Instructor – *Extreme Learning Center*, 2015–2017
- 3D Modeling & Printing Instructor – *Galileo*, 2015
- Editing Director & Camera Support – *Rainbow TV*, 2014–2015
- Journalist & Art Director – *EyeCandy Film Journal*, 2014
- Volunteer Math Tutor – *Live Oak Family Resource Center*, 2014
- Rotoscoper – *Short Film, Last Day of Freedom*, 2013

AWARDS & RECOGNITION

- **1st Place, AI for Good Hackathon** – AniTech, for “Fetch”, 2026
- **Achieve Award** – Nordson Medical, 2023
- **Inspire Award** – Nordson Medical, 2022
- **Biomedical Engineering Program Service Award** – SJSU, 2022
- **Research, Scholarship, and Creative Activity Grant** – SJSU, 2021
- **ZinnStarter Accelerator Fellow** – AniTech, for “The K-Tree”, 2021–2022
- **Best Product Innovation** – Silicon Valley Innovation Challenge, for “The K-Tree”, 2021
- **UC Berkeley TTE Research Certificate**, 2018
- **Porter College Service Award** for “ENTITLED”, Art Exhibition– UCSC, 2015
- **Porter College Undergraduate Fellowships** – UCSC, March & May 2015
- **Creative & Innovative Grant** – UCSC, 2015
- **Porter College Fellowships in Film & Digital Media** – UCSC, February & May 2014

SKILLS

Hard Skills
SolidWorks (CSWA), Autodesk Inventor, AutoCAD, Blender, 3D Printing, Arduino, Instron Testing, Microscopy, FIJI/ImageJ, RStudio, JASP, Minitab, Adobe Creative Suite, Canva, Wix, Filmmaking, Photography, Copywriting

Soft Skills
Clinical Need Identification, Human-Centered Design, Rapid Prototyping, Grant Writing, Team Leadership, Cross-Disciplinary Collaboration, Communication, Creative Problem-Solving, Project Coordination